

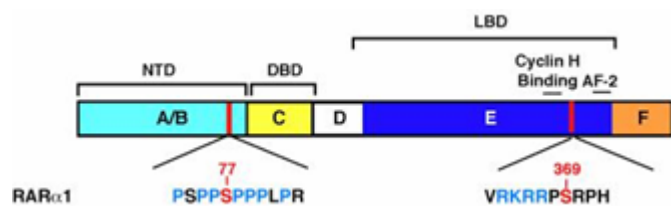
Genomics in Molecular Medicine Bioinformatics Coursework by UP89480 Part 1 (Dr Sharpe)

Introduction

Dynamic and combinatorial control of gene expression by nuclear retinoic acid receptors Rochette-Egly and Germain, (2009) Nuclear Receptor Signaling **7** (PMID:19471584)

The RA-activated p38MAPK/MSK1 pathway leads to a coordinated phosphorylation cascade targeting the receptor RAR α , its coregulators and associated histones. Recent studies from our laboratory demonstrated that RAR α becomes rapidly phosphorylated in response to RA at two serine residues, one being located in the LBD (ligand binding domain)(serine 369) and the other one in the NTD (N-terminal domain)(serine 77) [Bruck et al., 2009] (see figure below). Interestingly, serine 369 is an exposed residue located between helices 9 and 10 within the LBD. It belongs to an arginine-lysine-rich motif that corresponds to a consensus phosphorylation motif for several kinases including the cyclic AMP-dependent protein kinase (PKA) and MSK1, suggesting that it might integrate several signaling pathways [Bruck et al., 2009; Gaillard et al., 2006]. In contrast, serine 77 located in the NTD, belongs to a proline-rich motif. The kinase responsible for phosphorylating this site has been identified as cdk7 [Rochette-Egly et al., 1997], the activity of which depends on its association with cyclin H and MAT1 to form the ternary cyclin-dependent kinase (CDK)-activating kinase (CAK) complex of TFIIF, a general transcription factor composed of 10 subunits [Giglia-Mari et al., 2004]. The correct positioning of the cdk7 kinase and thereby the efficiency of the NTD phosphorylation by cdk7 rely on the docking of cyclin H at a specific site of the LBD located in L8-9 and the N-terminal part of H9 [Bour et al., 2005a; Gaillard et al., 2006]. To our knowledge, it was the first example of cooperation between the N- and C-terminal domains of RARs through a kinase complex. The docking site of cyclin H is conserved between RARs, but not in other nuclear receptors and other cdk7 targets, and differs from the other cyclin docking sites. Therefore, it would elicit the specificity of RAR phosphorylation by cdk7.

Recently, Bruck et al. demonstrated that the RA-induced phosphorylation of RAR α results from a coordinated phosphorylation cascade starting with the phosphorylation of serine 369 by MSK1 [Bruck et al., 2009]. As serine 369 is in close proximity to the docking site for cyclin H, phosphorylation of this residue induces an allosteric network, which promotes the binding of cyclin H [Gaillard et al., 2006] and thereby of cdk7, with a characteristic downstream consequence on the phosphorylation of the NTD at serine 77. In the following questions use RAR alpha transcript variant 2 RARA mRNA NM_001024809



1. The RAR alpha protein is activated by phosphorylation at two sites (in red) as described above

Use the NCBI site (Resources -> Gene -> RARA) to collect the RARA **protein** sequences for a *representative* range of up to 10 vertebrates from the bony fish to mammals. You do not need to show the FASTA files, but justify your selection of species **10%**

The 10 vertebrates chosen were: *Homo sapiens*, *Mus musculus*, *Danio rerio*, *Gallus gallus*, *Bos taurus*, *Canis lupus familiaris*, *Gorilla gorilla*, *Pan troglodytes*, *Xenopus tropicalis* and *Phocoena sinus*

Vaquita was chosen as it had the largest residue count of 583. Therefore, it is interesting to see what deletions and insertions are present compared to other species. The Western Gorilla and the Chimpanzee were chosen as they both species are genetically similar to humans and all belong to the Hominidae family. *Xenopus tropicalis* was chosen as one of the six primary model organisms used in development and it shares a similar number of residues at 463 compared to humans at 462 residues. *Mus musculus*, cattle and dog were chosen as all three species share the number of residues at 462 residues. Zebrafish was also chosen as it is one of the six primary model organisms. Also, Zebrafish has 447 amino acids compared to humans which have 462. Chicken was chosen as it was of its high

2. Using the Clustal Omega site, present an annotated alignment of the RARA sequences from the selected sequences **10%**

XP_Zebrafish_RARA	-----MSGKGNPSPHLNGFPMAHFSYFFPHML-----GGLSPPS-----LPTLPISGYS	44
XP_Xenopustropicalis_RARA	MTSNDNTGPPSGPGHMGNYVHPHYAFFFPHML-----GGMSPTGGLPGVGQHLPTISGYS	54
NP_Homo_sapiens_RARA	MY----ESVEVGGETPNPFLLVV---DF-YQNRACLLPKGLPAGPYSTP---L-RTPLW	49
NP_Chicken_RARA	MASNSSSCPTPGGGHLNGYPVTPTYAFFFPHML-----GGLSPSSLPGIQHQLPVSGYS	54
XP_Musmusculus_RARA	MASNSSSCPTPGGGHLNGYPVPFYAFFFPML-----GGLSPPGALTSLQHQLPVSGYS	54
NP_Dog_RARA	MASNSSSCPTPGGGHLNGYPVPFYAFFFPML-----GGLSPPGALTTLQHQLPVSGYS	54
XP_Phocoenasinus_RARA	MASNSSSCPTPGGGHLNGYPVPFYAFFFPML-----GGLSPPGALTTLQHQLPVSGYS	54
XP_Cattle_RARA	MASNSSSCPTPGGGHLNGYPVPFYAFFFPML-----GGLSPPGALTTLQHQLPVSGYS	54
XP_Chimpanzee_RARA	MASNSSSCPTPGGGHLNGYPVPFYAFFFPML-----GGLSPPGALTTLQHQLPVSGYS	54
XP_Gorilla_RARA	MASNSSSCPTPGGGHLNGYPVPFYAFFFPML-----GGLSPPGALTTLQHQLPVSGYS	54
	* : : : * : *	
	Cys65-Cys68 Ser77 Cys95-Cys96	
XP_Zebrafish_RARA	TPSPA--TIEQTSSSEIIVPSPPSPPPLEIRIYKPCFVCQDKSSGYHYGVSAACEGKGFF	102
XP_Xenopustropicalis_RARA	TPSPA KSIETQ TSSSEIIVPSPPSPPPLEIRIYKPCFVCQDKSSGYHYGVSAACEGKGFF	114
NP_Homo_sapiens_RARA	NGSNH--SIEQTSSSEIIVPSPPSPPPLEIRIYKPCFVCQDKSSGYHYGVSAACEGKGFF	107
NP_Chicken_RARA	TPSPA--TVETQSTSSSEIIVPSPPSPPPLEIRIYKPCFVCQDKSSGYHYGVSAACEGKGFF	112
XP_Musmusculus_RARA	TPSPA--TIEQTSSSEIIVPSPPSPPPLEIRIYKPCFVCQDKSSGYHYGVSAACEGKGFF	112
NP_Dog_RARA	TPSPA--TIEQTSSSEIIVPSPPSPPPLEIRIYKPCFVCQDKSSGYHYGVSAACEGKGFF	112
XP_Phocoenasinus_RARA	TPSPA--TIEQTSSSEIIVPSPPSPPPLEIRIYKPCFVCQDKSSGYHYGVSAACEGKGFF	112
XP_Cattle_RARA	TPSPA--TIEQTSSSEIIVPSPPSPPPLEIRIYKPCFVCQDKSSGYHYGVSAACEGKGFF	112
XP_Chimpanzee_RARA	TPSPA--TIEQTSSSEIIVPSPPSPPPLEIRIYKPCFVCQDKSSGYHYGVSAACEGKGFF	112
XP_Gorilla_RARA	TPSPA--TIEQTSSSEIIVPSPPSPPPLEIRIYKPCFVCQDKSSGYHYGVSAACEGKGFF	112
	. * ::***:***** **	
	Proline Rich Motif 73-82	
XP_Zebrafish_RARA	RRSIQKNMVYTCHREKNCIINKVTRNRCQYCRLOKCFEVGMSKESVRNDRNKKKEEKKP	162
XP_Xenopustropicalis_RARA	RRSIQKNMVYTCHRDKNCIINKVTRNRCQYCRLOKCFEVGMSKESVRNDRNKKKEBPKA	174
NP_Homo_sapiens_RARA	RRSIQKNMVYTCHRDKNCIINKVTRNRCQYCRLOKCFEVGMSKESVRNDRNKKKEVPKP	167
NP_Chicken_RARA	RRSIQKNMVYTCHRDKNCIINKVTRNRCQYCRLOKCFEVGMSKESVRNDRNKKKDVPKT	172
XP_Musmusculus_RARA	RRSIQKNMVYTCHRDKNCIINKVTRNRCQYCRLOKCFDVGMSKESVRNDRNKKKEAPKP	172
NP_Dog_RARA	RRSIQKNMVYTCHRDKNCIINKVTRNRCQYCRLOKCFEVGMSKESVRNDRNKKKEAPKP	172
XP_Phocoenasinus_RARA	RRSIQKNMVYTCHRDKNCIINKVTRNRCQYCRLOKCFEVGMSRESVRNDRNKKKEAPKP	172
XP_Cattle_RARA	RRSIQKNMVYTCHRDKNCIINKVTRNRCQYCRLOKCFEVGMSKESVRNDRNKKKEVPKP	172
XP_Chimpanzee_RARA	RRSIQKNMVYTCHRDKNCIINKVTRNRCQYCRLOKCFEVGMSKESVRNDRNKKKEVPKP	172
XP_Gorilla_RARA	RRSIQKNMVYTCHRDKNCIINKVTRNRCQYCRLOKCFEVGMSKESVRNDRNKKKEVPKP	172
	*****::*****:*****:	
	Cys207-208 Basic Region 154-162	
XP_Zebrafish_RARA	ECTENYTLSPDTEQMIDRVRKHAHQETFPALCQLGKYTTNSSEERRVALDLWDKFSELS	222
XP_Xenopustropicalis_RARA	DVMESYVLSPEITEDLIKVRKAHQETFPALCQLGKYTTNFSEERSVLIDLWDKFSELS	234
NP_Homo_sapiens_RARA	ECSESYTLTPEVGELIEKVRKAHQETFPALCQLGKYTTNNSSSEQRVSIDLWDKFSELS	227
NP_Chicken_RARA	ECSESYIVTPEVEELIKVRKAHQETFPALCQLGKYTTNNSSSEQRVSIDLWDKFSELS	232
XP_Musmusculus_RARA	ECSESYTLTPEVGELIEKVRKAHQETFPALCQLGKYTTNNSSSEQRVSIDLWDKFSELS	232
NP_Dog_RARA	ECSESYTLTPEVGELIEKVRKAHQETFPALCQLGKYTTNNSSSEQRVSIDLWDKFSELS	232
XP_Phocoenasinus_RARA	ECSESYTLTPEVGELIEKVRKAHQETFPALCQLGKYTTNNSSSEQRVSIDLWDKFSELS	232
XP_Cattle_RARA	ECSESYTLTPEVGELIEKVRKAHQETFPALCQLGKYTTNNSSSEQRVSIDLWDKFSELS	232
XP_Chimpanzee_RARA	ECSESYTLTPEVGELIEKVRKAHQETFPALCQLGKYTTNNSSSEQRVSIDLWDKFSELS	232
XP_Gorilla_RARA	ECSESYTLTPEVGELIEKVRKAHQETFPALCQLGKYTTNNSSSEQRVSIDLWDKFSELS	232
	: * *:*****:*****: ***:*****:	

XP_Zebrafish_RARA	TKCIKKTVEFAKQLPGFTTLTIADQITLLKAACLDILILRICTRYTPEQDTMTFSDGLTL	282
XP_Xenopustropicalis_RARA	TKCIKKTVEFAKQLPGFTTLTIADQITLLKSACLDILILRICTRYTPEQDTMTFSDGLTL	294
NP_Homo_sapiens_RARA	TKCIKKTVEFAKQLPGFTTLTIADQITLLKAACLDILILRICTRYTPEQDTMTFSDGLTL	287
NP_Chicken_RARA	TKCIKKTVEFAKQLPGFTTLTIADQITLLKAACLDILILRICTRYTPEQDTMTFSDGLTL	292
XP_Musmusculus_RARA	TKCIKKTVEFAKQLPGFTTLTIADQITLLKAACLDILILRICTRYTPEQDTMTFSDGLTL	292
NP_Dog_RARA	TKCIKKTVEFAKQLPGFTTLTIADQITLLKAACLDILILRICTRYTPEQDTMTFSEGLTL	292
XP_Phocoenasinus_RARA	TKCIKKTVEFAKQLPGFTTLTIADQITLLKAACLDILILRICTRYTPEQDTMTFSDGLTL	292
XP_Cattle_RARA	TKCIKKTVEFAKQLPGFTTLTIADQITLLKAACLDILILRICTRYTPEQDTMTFSDGLTL	292
XP_Chimpanzee_RARA	TKCIKKTVEFAKQLPGFTTLTIADQITLLKAACLDILILRICTRYTPEQDTMTFSDGLTL	292
XP_Gorilla_RARA	TKCIKKTVEFAKQLPGFTTLTIADQITLLKAACLDILILRICTRYTPEQDTMTFSDGLTL	292
*****:*****:*****:***		
XP_Zebrafish_RARA	NRTQMHNAGFGPLTDLVFAFANQLLPLEMDDAETGLLSAICLICGDRQDLEQADKVDVLQ	342
XP_Xenopustropicalis_RARA	NRTQMHNAGFGPLTDLVFAFANQLLPLEMDDAETGLLSAICLICGDRQDLEQPDKVDKLQ	354
NP_Homo_sapiens_RARA	NRTQMHNAGFGPLTDLVFAFANQLLPLEMDDAETGLLSAICLICGDRQDLEQPDPRVDMQLQ	347
NP_Chicken_RARA	NRTQMHNAGFGPLTDLVFAFANQLLPLEMDDAETGLLSAICLICGDRQDLEQPDKVDKLQ	352
XP_Musmusculus_RARA	NRTQMHNAGFGPLTDLVFAFANQLLPLEMDDAETGLLSAICLICGDRQDLEQPDKVDMLQ	352
NP_Dog_RARA	NRTQMHNAGFGPLTDLVFAFANQLLPLEMDDAETGLLSAICLICGDRQDLEQPDPRVDMQLQ	352
XP_Phocoenasinus_RARA	NRTQMHNAGFGPLTDLVFAFANQLLPLEMDDAETGLLSAICLICGDRQDLEQPDPRVDMQLQ	352
XP_Cattle_RARA	NRTQMHNAGFGPLTDLVFAFANQLLPLEMDDAETGLLSAICLICGDRQDLEQPDPRVDMQLQ	352
XP_Chimpanzee_RARA	NRTQMHNAGFGPLTDLVFAFANQLLPLEMDDAETGLLSAICLICGDRQDLEQPDPRVDMQLQ	352
XP_Gorilla_RARA	NRTQMHNAGFGPLTDLVFAFANQLLPLEMDDAETGLLSAICLICGDRQDLEQPDPRVDMQLQ	352
*****:*****:*****:***		
Ser369		
XP_Zebrafish_RARA	EPLLEALKIYVVRNRPPKPHMFPKMLMKITDLRSISAKGAERVITLKMEIPGSMPLIQE	402
XP_Xenopustropicalis_RARA	EPLLEALKIYVVRNRPPKPHMFPKMLMKITDLRSISAKGAERVITLKMEIPGSMPLIQE	414
NP_Homo_sapiens_RARA	EPLLEALKIYVVRNRPPKPHMFPKMLMKITDLRSISAKGAERVITLKMEIPGSMPLIQE	407
NP_Chicken_RARA	EPLLEALKIYVVRNRPPKPHMFPKMLMKITDLRSISAKGAERVITLKMEIPGSMPLIQE	412
XP_Musmusculus_RARA	EPLLEALKIYVVRNRPPKPHMFPKMLMKITDLRSISAKGAERVITLKMEIPGSMPLIQE	412
NP_Dog_RARA	EPLLEALKIYVVRNRPPKPHMFPKMLMKITDLRSISAKGAERVITLKMEIPGSMPLIQE	412
XP_Phocoenasinus_RARA	EPLLEALKIYVVRNRPPKPHMFPKMLMKITDLRSISAKGAERVITLKMEIPGSMPLIQE	412
XP_Cattle_RARA	EPLLEALKIYVVRNRPPKPHMFPKMLMKITDLRSISAKGAERVITLKMEIPGSMPLIQE	412
XP_Chimpanzee_RARA	EPLLEALKIYVVRNRPPKPHMFPKMLMKITDLRSISAKGAERVITLKMEIPGSMPLIQE	412
XP_Gorilla_RARA	EPLLEALKIYVVRNRPPKPHMFPKMLMKITDLRSISAKGAERVITLKMEIPGSMPLIQE	412
*****:*****:*****:***		
Basic Region 360-367 Cys 446-447 Cys 450-451		
XP_Zebrafish_RARA	MLENSEGLESSSGAQGSR-----ASATTPGSCSPSLSPNSAQSSPPTQSP	447
XP_Xenopustropicalis_RARA	MLENSEGLDTLGGGGGASGG-APPTVPVPPRSCSPSLSPSSTHSSPSTHSP	463
NP_Homo_sapiens_RARA	MLENSEGLDTLGSGQPGGGGRDGGGLAPPPGSCSPSLSPSSNRSSPATHSP	457
NP_Chicken_RARA	MLENSEGMDTLGGQPGGP--RTGGLGPPPGSCSPSLSPSSTRSSPATHSP	460
XP_Musmusculus_RARA	MLENSEGLDTLGSGQSGGGTRDGGGLAPPPGSCSPSLSPSSHRSSPATQSP	462
NP_Dog_RARA	MLENSEGLDTLGSGQPGGGGRDGGGLAPPPGSCSPSLSPSSNRSSPATHSP	462
XP_Phocoenasinus_RARA	MLENSEGLDTLGSGQAGGGGRDGGGLAPPPGSCSPSLSPSSNRSSPATHSP	462
XP_Cattle_RARA	MLENSEGLDTLGSGQAGSGGRDGGGLAPPPGSCSPSLSPSSNRSSPATHSP	462
XP_Chimpanzee_RARA	MLENSEGLDTLGSGQPGGGGRDGGGLAPPPGSCSPSLSPSSNRSSPATHSP	462
XP_Gorilla_RARA	MLENSEGLDTLGSGQPGGGGRDGGGLAPPPGSCSPSLSPSSNRSSPATHSP	462
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Figure 1: Alignment of RARA between various species: human, *Xenopus tropicalis*, chicken, mouse, Dog, cattle, gorilla, zebrafish, *Phocoena sinus*, chimpanzee. With the cystine bridges marked in red, the basic regions marked in orange, the proline rich region marked in magenta and the serine phosphorylation sites are marked in blue.

- Based on this data, generate and present a simple phylogenetic tree of RARA sequences and then mark which of the two serine residues relevant to protein activation are present in each species. (You may need to download the Clustal tree image and post-annotate it in powerpoint, word or an equivalent app) **20%**

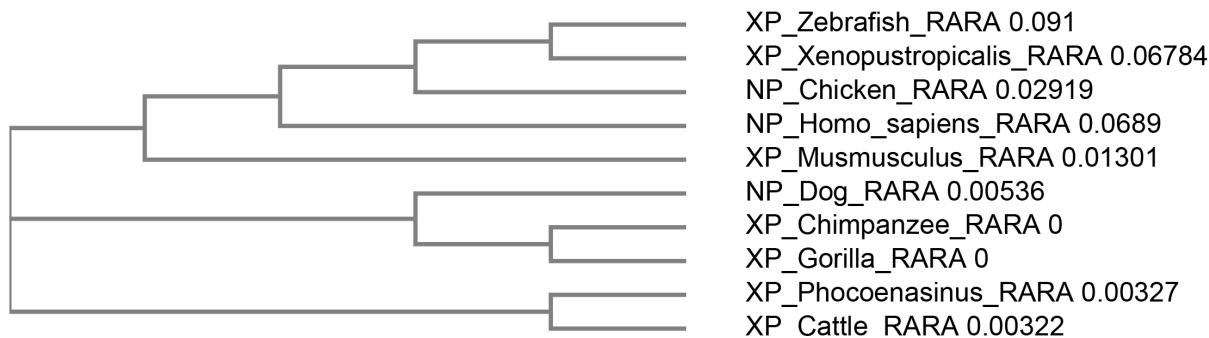


Figure 2: A phylogenetic tree of RARA between various species: human, *Xenopus tropicalis*, chicken, mouse, Dog, cattle, gorilla, zebrafish, *Phocoena sinus*, chimpanzee.

The phylogenetic tree of RARA sequences displays chimpanzee and Gorilla RARA sequences are highly conserved amongst each other. Cattle and vaquita sequences are similar in their RARA sequence despite belonging to different families. Humans and Apes belong to the same family however human RARA is dissimilar to the apes but more genetically similar to Chicken and Mouse. Zebrafish and *Xenopus Tropicalis* RARA are the most genetically distant in the series.

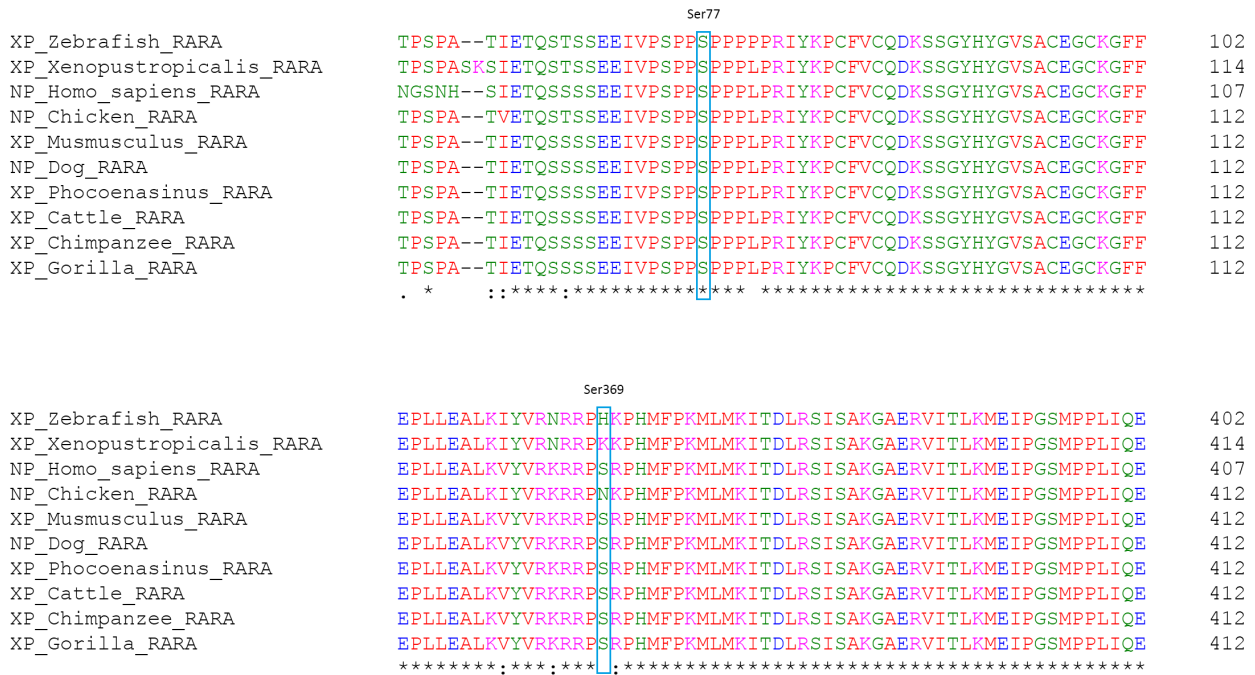


Figure 3: Alignment of RARA between various species. From top to bottom, Zebrafish, *X.tropicalis*, human, chicken, dog, vaquita, cattle, chimpanzee and gorilla. Serine 77 and serine 369 are highlighted in blue.

The clustal alignment of RARA sequences display trends amongst the different species from. All of the RARA sequences displayed are highly conserved amongst each other. All species possess the same Ser77 and Ser369 phosphorylation sites.

- Using Blast2seq, compare the human RARA protein sequence with its paralogue RARB. Is it likely that RAR beta is also activated by a similar mechanism of two site phosphorylation? Present alignments to support your conclusion **20%**

RARA	1	MYESVEVGGPTPNPFLVDFYNQN-RACLLPEKGLPAPGPYSTPLRTPPLWNGSN--HSIE	57
		M++ ++V +P L DFY + +C+L EK L A +S +T W + SIE	
RARB	1	MFDCMDVLSVSPGQIL--DFYTASPSSCMLQEKAALKAC--FSGLTQTE-WQHRHTAQSI	55
RARA	58	TQSSSSEEIVSPSPSPPLPRIYKPCFVCQDKSSGYHYGVSACEGCKGFFRRSIQKNMVY	117
		TQS+SSEE+VPSPPSP P PR+YKPCFVCQDKSSGYHYGVSACEGCKGFFRRSIQKNM+Y	
RARB	56	TQSTSSEELVSPSPSPPLPPRVYKPCFVCQDKSSGYHYGVSACEGCKGFFRRSIQKNMIY	115
RARA	118	TCHRDKNCIINKVTRNRCQYCRLOKCFEVGMSKESVRNDRNKKKKEVPKPECSESYTLTP	177
		TCHRDKNC+INKVTRNRCQYCRLOKCFEVGMSKESVRNDRNKKKKE K EC+ESY +T	
RARB	116	TCHRDKNCVINKVTRNRCQYCRLOKCFEVGMSKESVRNDRNKKKKEKTSKQECTESYEMTA	175
RARA	178	EVGELIEKVRKAHQETFPALCQLGKYTTNNSSEQRVSLDIDLWDFSELSTKCIKTVEF	237
		E+ +L EK+RKAHQETFP+LCQLGKYTTN+S++ RV LD+ LWDKFSEL+TKCIK VEF	
RARB	176	ELDDLTEKIRKAHQETFPSPSLCQLGKYTTNNSADHRVRLDLGLWDFSELATKCIKIVEF	235
RARA	238	AKQLPGFTTLTIADQITLLKAACLDILILRICTRYTPEQDTMTFSDGLTLNRTQMHNAGF	297
		AK+LPGFT LTIADQITLLKAACLDILILRICTRYTPEQDTMTFSDGLTLNRTQMHNAGF	
RARB	236	AKRLPGFTGLTIADQITLLKAACLDILILRICTRYTPEQDTMTFSDGLTLNRTQMHNAGF	295
RARA	298	GPLTDLVFAFANQQLPLEMDDAETGLLSAICLICGDRQDLEQPDRLVDMLEPLLEALKVY	357
		GPLTDLVF FANQQLPLEMDD ETGLLSAICLICGDRQDLE+P +VD LQEPLLEALK+Y	
RARB	296	GPLTDLVTFANQQLPLEMDDTETGLLSAICLICGDRQDLEETPKVDKLQEPLLEALKIY	355
RARA	358	VRKRRPSRPHMFPKMLMKITDLRSISAKGAERVITLKMEIPGSMPLIQEMLENSEGLDT	417
		+RKRPS+PHMFPK+LMKITDLRSISAKGAERVITLKMEIPGSMPLIQEMLENSEG +	
RARB	356	IRKRRPSKPHMFPKILMKITDLRSISAKGAERVITLKMEIPGSMPLIQEMLENSEGHEP	415
RARA	418	LSGQPGGGGRDGGGLAPPPGSCSPSLSPSSNRSSPATHSP	457
		L+ G + SPS+SPSS +S + SP	
RARB	416	LTPSSSGNTAEH-----SPSISPSSSVENSGVSQSP	445

Figure 4: Alignment of human RARA and human RARB with the serine phosphorylation residues highlighted in orange.

Human RARA protein sequence with its paralogue RARB both share the basic arginine-lysine-rich motif which have Ser369 located. Likewise Ser77, highlighted in orange. Found in the proline-rich motif which is shared between RARA and RARB. The sequence homologies between the two proteins is 78.5%, concluding that RARB is activated by a similar mechanism of two site phosphorylation by kinases as found in RARA.

5. The disease, Duchenne Muscular Dystrophy is caused by mutations to the dystrophin gene (DMD). It is a very long gene and its exons tend to encode repeating domains in the protein. A mutation causing a deletion of one base pair in exon 52 will produce a frameshift in the remaining coding sequence and generate a pathogenic non-functional protein.

Present a translation of the mRNA encoded by exons 51/52/53/54 and compare to 51/52*/53/54 where exon 52* carries the point mutation. The sequences are below **10%**

Expassy Translate 1

> Wild type DMD exons 51-54 5'3' Frame 3

```
MPSSLMLEVPALADFNRAWTELTDWLSLLDQVIKSQRVMVGDLEDINEMI
IKQKATMQDLEQRRPQLEELITAAQNLKNKTSNQEARTIITDRIERIQNQ
WDEVQEHQLNRRQQLNEMLKDSTQWLEAKEEAEQVLGQARAKLESWKEGP
YTVDAIQKKITETK
```

> Single base deleted DMD exons 51-54 5'3' Frame 3

MPSSLMLEVPALADFNRAWTELTDWLSLLDQVIKSQRVMVGDLEDINEMI
IKQKATMQDLEQRRPSWKNSLPLPKI

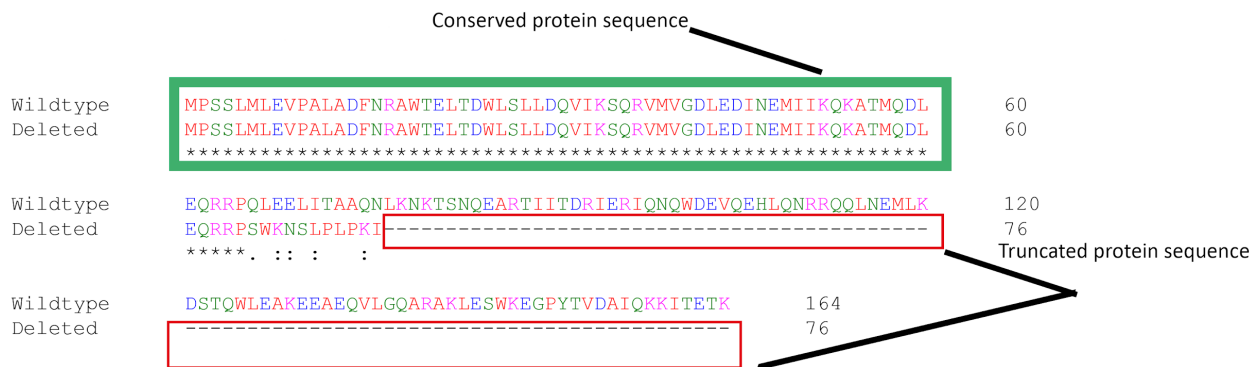


Figure 5: Alignment of wildtype DMD exons 51-54 5'3' Frame 3 and Single base deleted DMD exons 51-54 5'3' Frame 3.

From position 1 to 60, the wildtype and deleted version of DMD are conserved. After amino acid 76, the deleted version of DMD produced a premature truncated protein sequence.

A proposed gene therapy for this form of DMD is to generate mRNA with deletions of either exon 52, or both exons 52 and 53, using antisense oligonucleotides to inhibit splicing. Will either strategy be successful in generating an open reading frame in which the amino acid sequence of exon 54 (and beyond) is the same as the original protein? Present translations of the different mRNAs produced by this procedure in this region of the DMD gene to support your answer **30%**

Expassy Translate 2

> Wild type DMD exons 51-54 5'3' Frame 3

MPSSLMLEVPALADFNRAWTELTDWLSLLDQVIKSQRVMVGDLEDINEMI
IKQKATMQDLEQRRPQLEELITAAQNLKNKTSNQEARTIITDRIERIQNQ
WDEVQEHLQNRQQLNEMLKDSTQWLEAKEEAEQVLGQARAKLESWKEGP
YTVDIAIQKKITETK

> Single base deleted DMD exons 51 and 54 5'3' Frame 3

MPSSLMLEVPALADFNRAWTELTDWLSLLDQVIKSQRVMVGDLEDINEMI
IKQKATMQDLEQRRPSWKNSLPLPKI

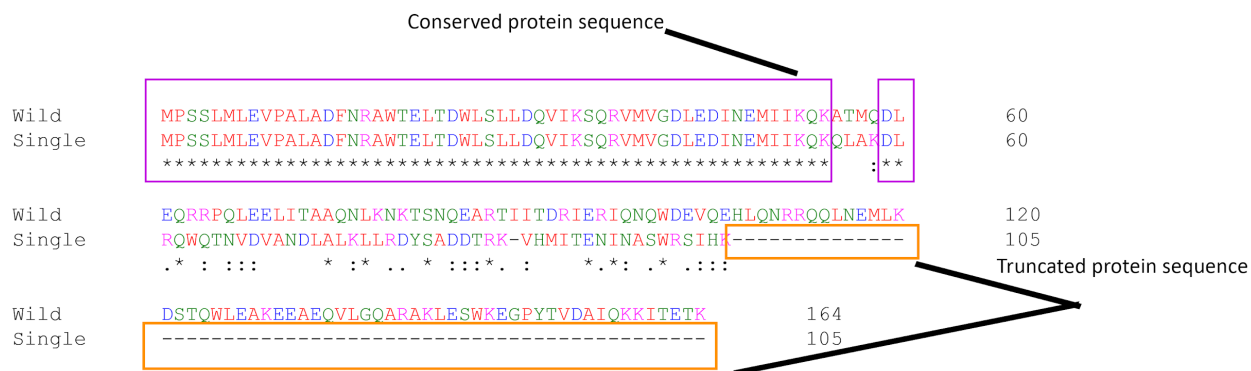


Figure 6: Alignment of wildtype DMD exons 51-54 5'3' Frame 3 and Single base deleted DMD exons 51 and 54 5'3' Frame 3 only.

After residue 60, the deleted DMD gene is partially recovered as the protein was recovered up to residue 105 where the deleted version of DMD with exon 52 and 53 removed produced a truncated protein sequence.

Expasy Translate 3

> Wild type DMD exons 51-54 5'3' Frame 3

```
MPSSLMLEVPALADFNRAWTELTDWLSLLDQVIKSQRVMVGDL
INEMI
IKQKATMQDL
EQRPPQLEELITAAQNLKNKTSNQEARTIITDRIERI
QWDEVQEH
LNRRQQLNEMLK
DSTQWLEAKEEAEQVLGQARAKLESWKEG
PYTVDAIQKKITETK
```

> Single base deleted DMD exons 51, 53 and 54 5'3' Frame 3

```
MPSSLMLEVPALADFNRAWTELTDWLSLLDQVIKSQRVMVGDL
INEMI
IKQKATMQDL
EQRPPQLEELITAAQNLKNKTSNQEARTIITDRIERI
QWDEVQEH
LNRRQQLNEMLK
DSTQWLEAKEEAEQVLGQARAKLESWKEG
PYTVDAIQKKITETK
```



Figure 7: Alignment of wildtype DMD exons 51-54 5'3' Frame 3 and Single base deleted DMD exons 51, 53 and 54 5'3' Frame 3 only.

After residue 54, the deleted DMD gene produces a non-homologous sequence. After residue 75, a truncated sequence is formed

Both varieties are different. The removal of both exon 52 and exon 53 generated an open reading frame and beyond exon 54 compared with the removal of only exon 52

Word Count: 515

Genomics in Molecular Medicine Bioinformatics Coursework by UP89480 Part 2 (Dr Bourgeois)

This part of the Bioinformatics assessment contributes **35% of the overall coursework mark**.

Here, we will apply our skills to identify the following protein sequence, and gather information about the gene that encodes it:

```
MAQQWSLQRLAGRHPQDSYEDSTQSSIFTYTNSNSTRGPFEGPNYHIAPRWVYHLTSVWM
IFVVTASVFTNGLVLAATMKFKKLRHPLNWILVNLAVADLAETVIASITISIVNQVSGYFV
LGHPMCVLEGYTVSLCGITGLWSLAIISWERWLTVCKPFGNVRFDKLAIVGIAFSWIWS
AVWTAPPIFGWSRYWPHGLKTSCGPDVFSGSSYPGVQSYMIVLMVTCCIIPLAIIMLCYL
QVWLAIRAVAKQKKESESTQKAEKEVTRMVVVMIFAYCVCWGPYTFFACFAAANPGYAFH
PLMAALPAYFAKSATIYNPVIYVFMNRQFRNCILQLFGKKVDDGSELSSASKTEVSSVSS
VSPA
```

1. Using the BLAT tool on Ensembl, identify the species and the gene to which this sequence belongs. **15%**

Homo Sapiens, OPN1LW

2. How many alternative transcripts are listed for this gene? **10%**

3 transcripts: OPN1LW-201 which has 1242 bp and protein-coding, OPN1LW-202 which has 592 bp and protein-coding. OPN1LW-203 which has 692 bp is not protein-coding.

3. Identify the three paralogs with the highest percentage of identity. What do you notice? **15%**

Three paralogs with the highest percentage are OPN1MW3, OPN1MW2 and OPN1MW1. Share the same percentage identity at 96.43% compared to the target sequence. The paralogs belong to the ancestral group of Hominoidea.

4. Do you observe ortholog genes in other vertebrates in *Drosophila melanogaster*? you can summarize your findings with a phylogeny **15%**. Feel free to comment on any aspect that you deem relevant.

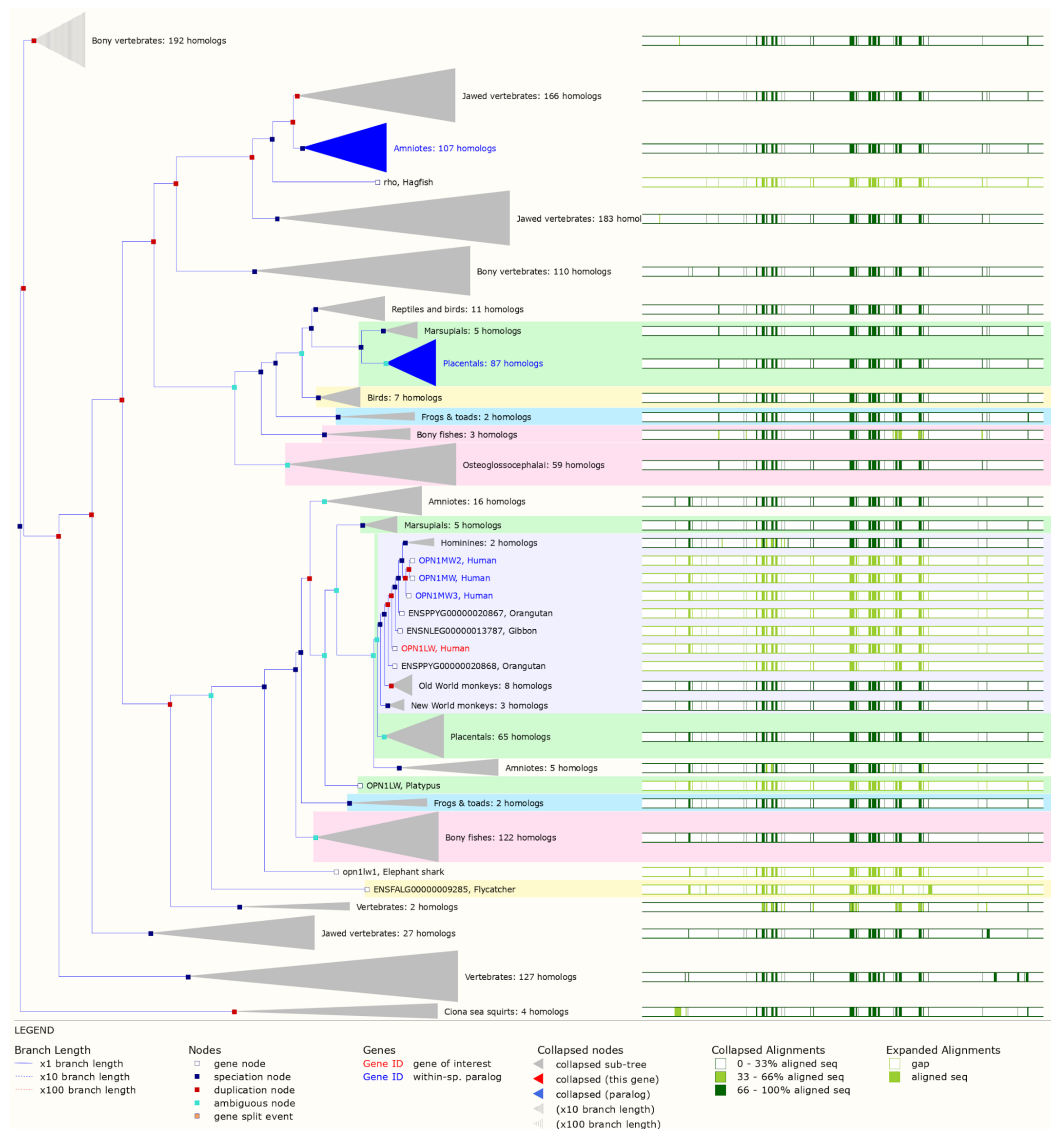


Figure 1: displays a phylogenetic tree for OPN1LW. All 741 chordates homologs are displayed as a speciation node. Human OPN1LW grouped along three paralogs, OPN1MW3, OPN1MW2 and OPN1MW1 alongside the apes.

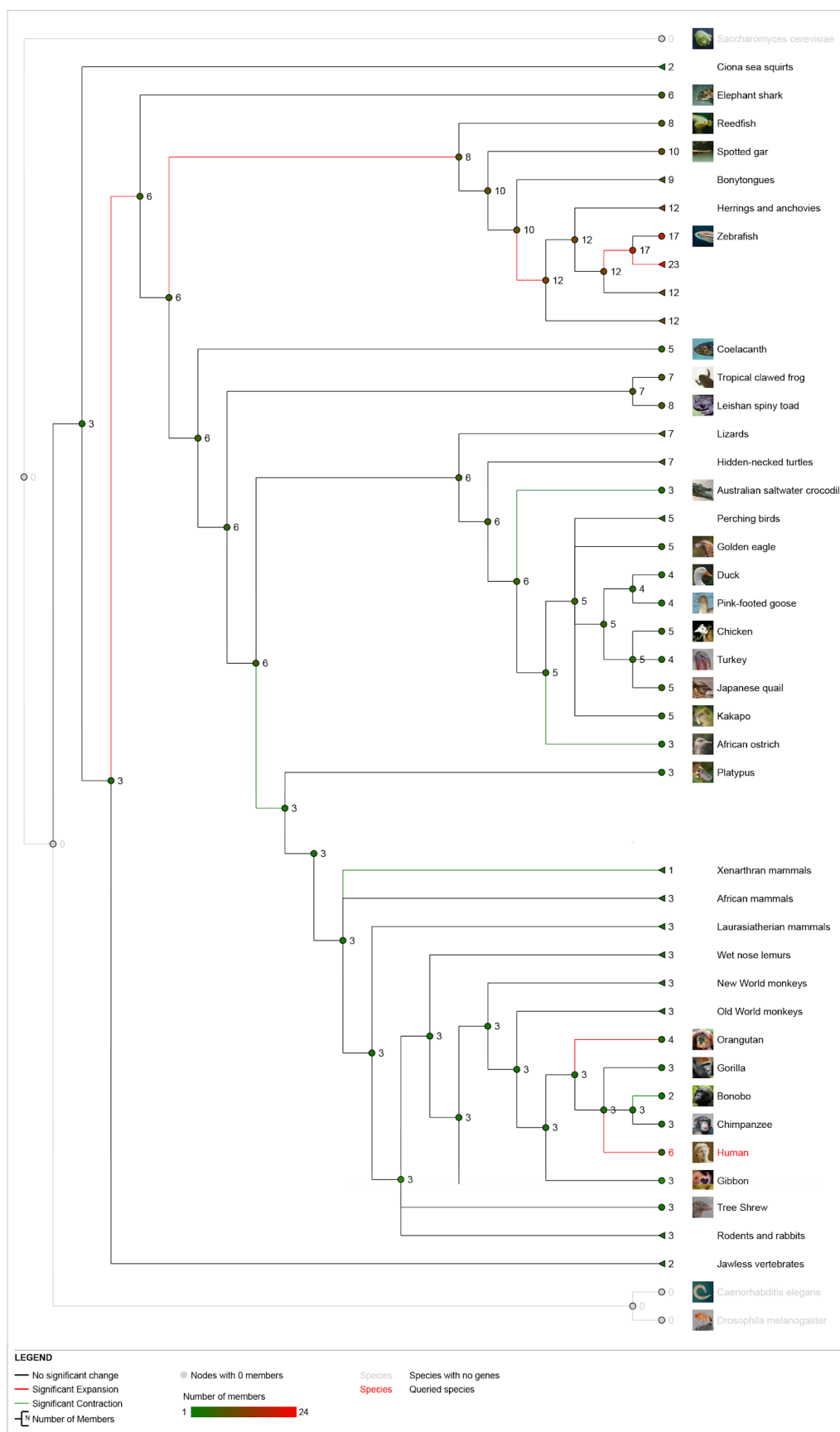


Figure 2: displays a gene gain/loss tree of *OPN1LW* across various species.

The ortholog genes are organised by taxonomic families. For example, the Hominidae are grouped in one part of the tree and Aves are found in another part of the tree. The bony fish which include reedfish to zebrafish are grouped whilst elephant shark remain separate. The saltwater crocodile version of *OPN1LW* is genetically similar to the family Aves. No ortholog genes in fruit fly were found.

- Using the UCSC genome browser, identify potential regulatory and promotor regions based on epigenetic marks and results from immune-precipitation experiments. **15%** You can justify your answer with Figures extracted from the browser.

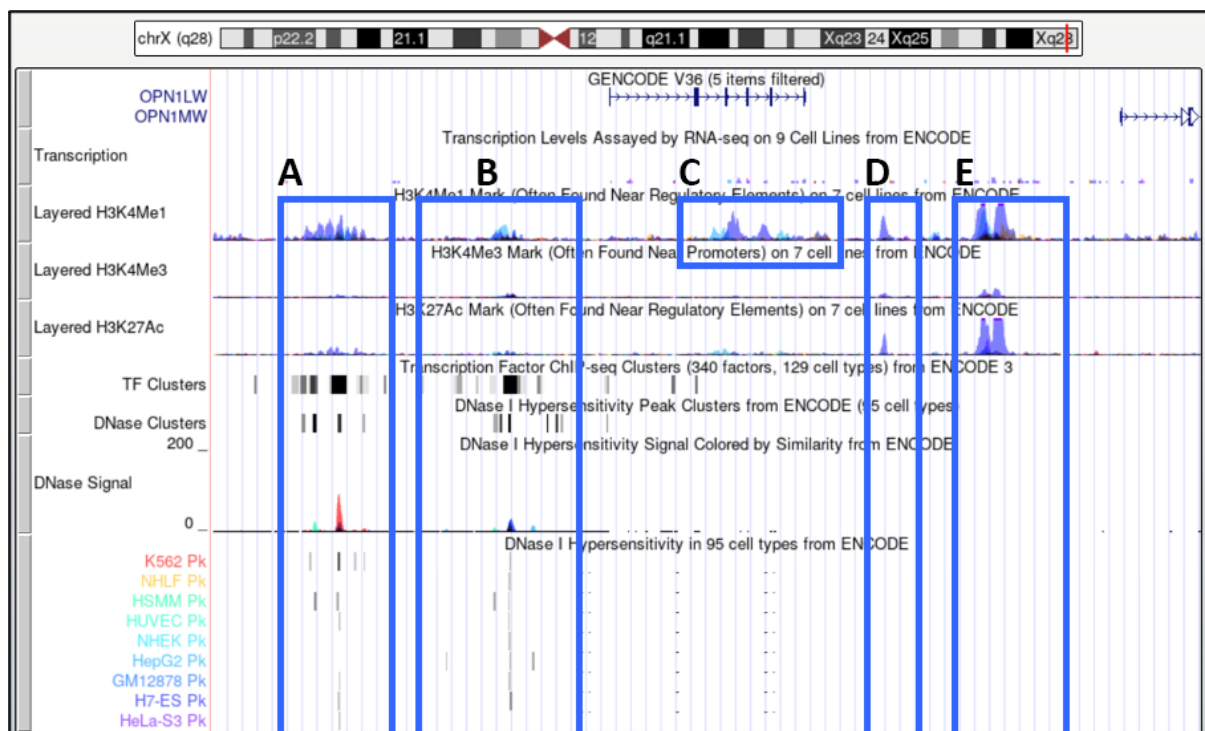


Figure 3: UCSC genome browser of the gene *OPN1LW* with the layered epigenetic marks and the immune-precipitation markers enabled.

Figure 3A displays high H3K4Me1 levels combined with transcription factors (TF) clusters which are proteins that bind to DNA and interact with RNA polymerase to regulate gene expression. The darker boxes, the likelihood of TFs binding in the region. The DNase clusters and DNase signals are found (Figure 3A, Figure 3B) means the chromatin has lost its heterochromatin structure, making it accessible for transcription. H3K4Me1 signatures are found near regulatory elements and mean the gene is poised and is not transcribed intermediately. The lack of H3K4Me3 signatures means they are likely promoters (Figure 3A, Figure 2B). Figure 3C show high H3K4Me1 levels implying they have a potential regulatory function. The two overlapping spikes for H3K4Me1 and H3K27Ac respectively (Figure 3D, Figure 3E). The lack of DNase clusters and TF clusters. Together with the acetylation which opens H3 chromatin means that it is associated with regulatory function.

6. Which phenotype(s) is (are) associated with variation at this gene in human populations? Summarize all your findings about this gene and its paralogs, and justify your answers with references to previous analyses **and scientific literature**. **30%**

Blue cone monochromatism which is a X-linked condition, which causes the loss of function in red cones and green cones (Blackwell & Blackwell, 1961; Nathans et al., 1989), leaving the blue cones and rods. Due to the loss of OPN1LW alongside OPN1MW and its regulated genes (Katagiri et al., 2018). Based on the earlier work of blue-cone monochromats from 12 families (Nathans et al., 1989) and further supported by the findings from Nathans et al., 1993 on 33 families identified two ways that blue-cone monochromacy can arise. The first way involves the unequal homologous recombination which reduces the number of visual pigment genes resulting in a point mutation, replacing Cys203 with Arg, breaking the disulphide bond (Nathans et al., 1993; Wang et al. 1992). The second way involves a nonhomologous deletion of DNA sequence adjacent to the red and green pigment clusters in the range of 3.1-3.7 kb and the point mutations from Arg247 to a terminal sequence in a single red-pigment and the other mutation results in Pro307 to Leu mutation in single 5' red-3' green hybrid gene (Nathans et al., 1993; Nathans et al., 1989; Wang et al. 1992).

Colour blindness, partial, protan series (CBP) is a type of colour blindness that affects long and medium cones encoded by OPN1LW and OPN1MW, arranged in a tandem array in X-chromosome at Xq28 (Nathans, Piantanida, Eddy, Shows & Hogness, 1986; Nathans, Thomas & Hogness, 1986; Neitz & Neitz, 2011; Vollrath, Nathans & Davis, 1988). This can be attributed to unequal homologous recombination during meiosis (Nathans, Piantanida, Eddy, Shows & Hogness, 1986). When X-chromosomes misalign, OPN1LW and OPN1MW are highly conserved sequences and perform crossover between paralogs. This produces a protan defect, which is a hybridisation between OPN1LW and OPN1MW and a deutan defect (Neitz & Neitz, 2011; Neitz, Neitz, He & Shevell, 1999; Shevell, He, Kainz, Neitz & Neitz, 1998)..

Cone-rod dystrophy results in the loss of rod and cone photoreceptors. The symptoms included decreased visibility, colour vision defects and photo-aversion (Hamel, 2007). The OPN1MW exon 3 sequence, a paralog to OPN1LW which contains a missense mutation that changes Trp177 to Arg177. Both share the same OPN1MW derived sequence of 492 bp and this allowing OPN1MW which contains missense mutation Trp177 to Arg177 to be transferred to OPN1LW (Gardner et al., 2011).

Word Count: 566

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Genomics in Molecular Medicine Bioinformatics Coursework by UP89480 Part 3 (Dr Bourgeois)

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>Gene_for_Coursework
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```

This part of the Bioinformatics assessment contributes **35% of the overall coursework mark**.

You will use Ensembl, Treefam, Genomicus and other databases to describe its evolution

1. Use NCBI BLAST and ENSEMBL to identify the gene of interest and the species. What can you tell about its function, its expression? **30%**

NCAM2, found in Homo sapiens. Belongs to the immunoglobulin superfamily of genes and encode for a type I transmembrane lipid-anchored protein and found in Chromosome 21. Its function is associated with the regulation of neurite growth the activation of FGF receptor and facilitate neuronal and axonal compartmentalisation. NCAM2 expression is found in the brain primarily but also in the adrenal glands, testis, liver and lungs ("NCAM2 neural cell adhesion molecule 2 [Homo sapiens]", n.d.; Rasmussen et al., 2018; Winther, Berezin & Walmod, 2012).

2. Using resources such as ENSEMBL and Genomicus, draw a diagram illustrating the conservation of synteny in five species. You will use **human, mouse, chicken, zebrafish**, plus one species of your choice. Show five genes before and after the gene of interest. Make sure to use several databases to confirm the homology of syntenic genes in the different species. Do not just draw a figure, and take the time to describe your results. **30%**

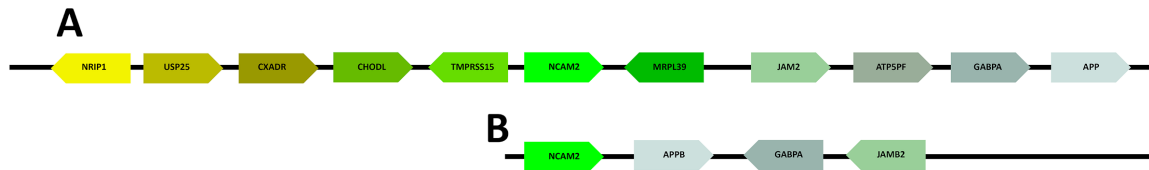


Figure 1: *A*, illustrating the conservation of synteny in human, mouse, chicken, and pig and *B*, represents zebrafish and the conservation of their synteny in comparison to *A*.

Between human, mouse, chicken, and pig there is a significant range of conservation of synteny the associated genes which neighbour the gene of interest, NCAM2. Zebrafish, however, (Figure 1B) only has three conserved genes from the neighbouring NCAM2: APPB, GABPA and JAMB2. APPB is a homolog to APP. The difference is that APPB is transcribed in a forward direction compared to APP which is transcribed in a reverse direction. JAM2B is another homolog to JAM2. Again the difference is that JAM2 is located in a forward directionality and the JAMB2 is located in a reverse directionality. GABPA is the only gene that shares near homology but the synteny between Figure 1A and Figure 1B are different. As in Figure GABPA is in a forward direction compare to Figure 1B in Zebrafish in a reverse direction. Genes such as USP25, CXADR and other genes from Figure 1A in Zebrafish are found in the genome but neighboured in a different set of genes.

3. Discuss the conservation of synteny in relation with the evolutionary history of vertebrates. For example, do you observe many inversions/deletions? How do these changes correlate with the phylogeny and why? **20%**

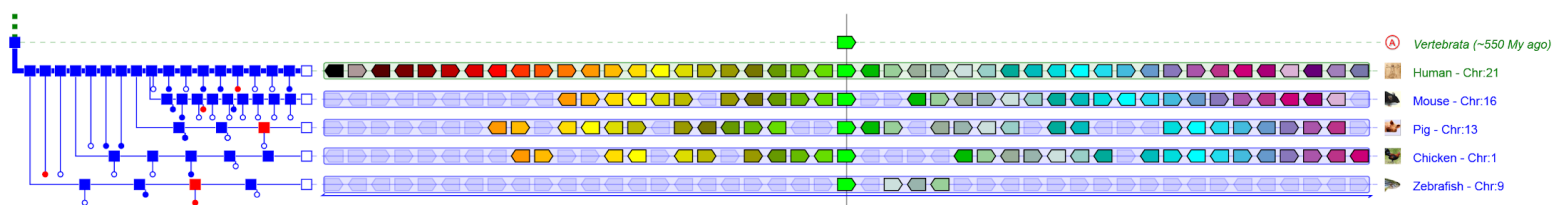


Figure 2: *Illustrating the conservation of synteny in human, mouse, chicken, pig and zebrafish represented in a schematic diagram.*

The conservation of synteny varies from species to species. The evolution diverges from the human sequence is conserved well in mice from the genes LIPI to KRTAP24 and has the fewest deletion in its sequence. From the human N6AMT1 which is found in reverse direction like in mice, pig and chicken which is in a forward direction. Chicken experiencing a greater expansion of its genes in the chromosome compared to its paralogs on the same chromosome. In zebrafish, there are two inversions GABPA and JAMB2 in its synteny whilst NCAM2 remains conserved and APPB is a paralog to APP and is found closer to NCAM2 with the rest of the genome is deleted. The difference between zebrafish and other vertebrates is due to several duplications. All vertebrates undergoing primary and secondary genome duplication (Dong, Fredman & Lenhard, 2009). Zebrafish which belongs to

teleost fish which underwent a third round of genome duplication (Lu, Peatman, Tang, Lewis & Liu, 2012) and undergone re-diploidisation by deletion of duplicated genes.

4. Identify genes that may be related to the gene of interest (e.g. paralogs) using for example Treefam. What do you observe? How do you interpret this observation given what you know of the evolution of vertebrates? **20%**

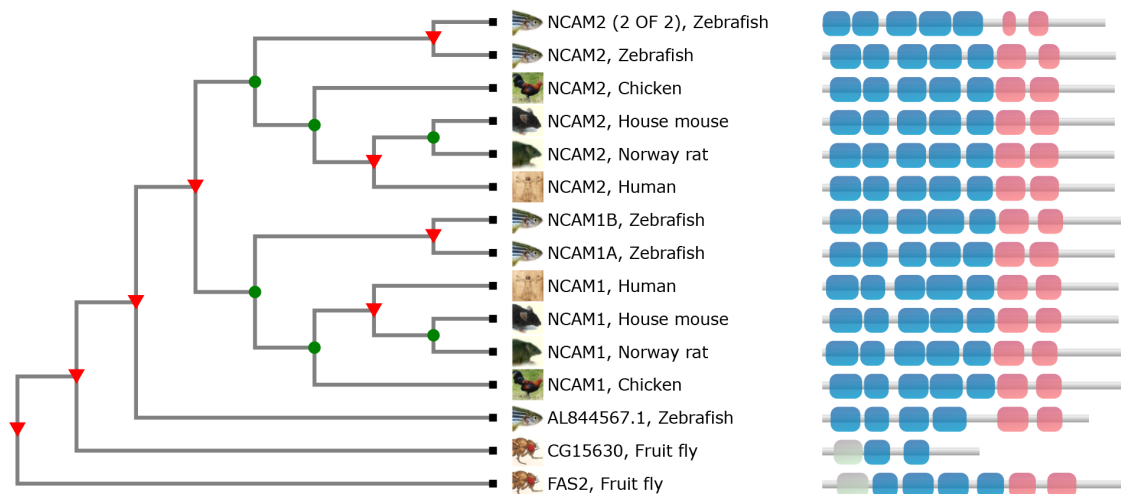


Figure 3: Illustrating a gene tree for NCAM2 between humans, chicken, zebrafish, fruit fly, the Norway rat and house mouse. Created in TreeFam.

In humans, NCAM2 has a paralog called NCAM1. The remaining vertebrates: chicken, the Norway rat, the house mouse and zebrafish. All have NCAM2 and NCAM1 homologs. This implies that the ancient NCAM of the ancestral vertebrate had undergone a genome duplication (Figure 3) and went through re-diploidisation by neo-functionalisation of the duplicated genes by splitting the function of the original NCAM into NCAM2 and NCAM1.

Zebrafish, there are three related orthologs that are related to the human NCAM2 gene. These are: AL844567.1, NCAM2 and NCAM1. NCAM2 in Zebrafish, the gene is split into NCAM2 and NCAM2 (2 OF 2) which are paralogs to each other and NCAM1 is split into NCAM1A and NCAM1B paralogs. This shows that the teleost fish such as the zebrafish underwent a third round of genome duplication.

The two genes found in Drosophila are FAS2 and CG15630, which are orthologous to human NCAM2. CG15630 has functions in cell projection and neuronal development (Fedotov et al., 2017) whilst FAS2 has the function of mediating homophilic adhesion (Neuert et al., 2019) associated with synaptic formation (Dityatev & Schachner, 2001; Sanes, Reh, Harris & Landgraf, 2019).

Word Count: 553

References

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Total Word Count for Genomics in Molecular Medicine Bioinformatics Coursework Part 1, 2 and 3:
1,634